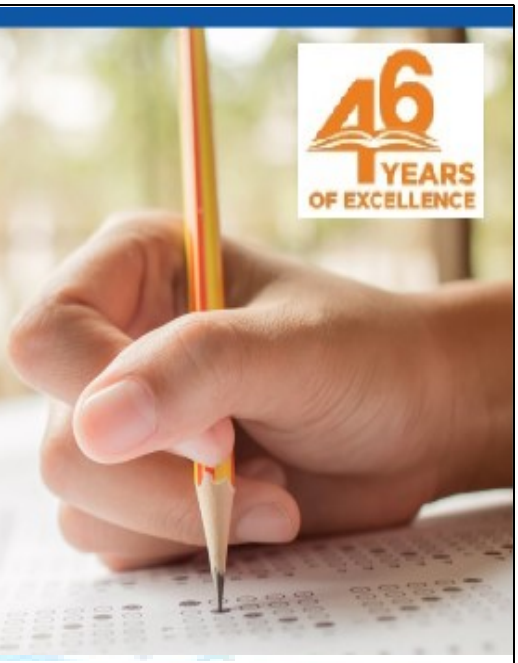
NARAYANA IIT ACADEMY INDIA

# JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 07-05-26
Time: 3 Hrs Max. Marks: 180

**07-05-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-12(P2)_SYLLABUS**

Rankers Academy JEE

**MATHEMATICS:** **TOTAL SYLLABUS**

**PHYSICS:** **TOTAL SYLLABUS**

**CHEMISTRY:** **TOTAL SYLLABUS**

THE NARAYANA GROUP

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**TIME: 3HRS** **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

## MATHEMATICS

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 1 – 4) | Questions with **Single Correct** Options | +3 | -1 | 4 | 12 |
| Sec – II (Q.N : 5 – 7) | Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0) | +4 | -2 | 3 | 12 |
| Sec – III (Q.N : 8 – 13) | Questions with **Non-Negative Integer** type | +4 | 0 | 6 | 24 |
| Sec – IV (Q.N : 14 – 17) | Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q) | +3 | 0 | 4 | 12 |
| **Total** | | | | **17** | **60** |

## PHYSICS

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 18 – 21) | Questions with **Single Correct** Options | +3 | -1 | 4 | 12 |
| Sec – II (Q.N : 22 – 24) | Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0) | +4 | -2 | 3 | 12 |
| Sec – III (Q.N : 25 – 30) | Questions with **Non-Negative Integer** type | +4 | 0 | 6 | 24 |
| Sec – IV (Q.N : 31 – 34) | Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q) | +3 | 0 | 4 | 12 |
| **Total** | | | | **17** | **60** |

## CHEMISTRY

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 35 – 38) | Questions with **Single Correct** Options | +3 | -1 | 4 | 12 |
| Sec – II (Q.N : 39 – 41) | Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0) | +4 | -2 | 3 | 12 |
| Sec – III (Q.N : 42 – 47) | Questions with **Non-Negative Integer** type | +4 | 0 | 6 | 24 |
| Sec – IV (Q.N : 48 –51) | Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q) | +3 | 0 | 4 | 12 |
| **Total** | | | | **17** | **60** |

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**MATHEMATICS** **MAX.MARKS: 60**

## SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks :** +3 If only the correct option is chosen.
**Zero Marks:** 0 If none of the option is chosen.(i.e the question is un answered)
**Negative Marks:** -1 In all other cases.

1. The coefficient of $x^{45}$ in $\prod_{i=1}^{10}\left(x^i - i\right)$ is:-

A) -8 B) -6 C) 6 D) 2

2. If $f: R \to R$ be a differentiable function such that $(f(x))^7 = x - f(x)$ then the area bounded by curve $y = f(x)$ between the ordinates $x = 0$ and $x = \sqrt{3}$ is

A) $\frac{f(\sqrt{3})}{8}\left[8\sqrt{3} - (f(\sqrt{3}))^7 - 4f(\sqrt{3})\right]$ sq. units

B) $\frac{f(\sqrt{3})}{4}\left[8\sqrt{3} - (f(\sqrt{3}))^7\right]$ sq. units

C) $\left(\sqrt{3}f(\sqrt{3}) - \frac{93}{8}\right)$ sq. units

D) $\frac{f(\sqrt{3})}{8}\left[4\sqrt{3} - (f(\sqrt{3}))^8 - 4f(\sqrt{3})\right]$ sq. units

3. If probability that $2^3 3^5 5^4 7^2 11^\alpha$ be resolved as product of two factors such that their highest common factor is 1 or 5 or 7 is $\left(\frac{1}{27}\right)$, then $\alpha$ is

A) 1 B) 2 C) 5 D) 4

4. Let $f:(0,\pi) \to R$ be a twice differentiable function such that $\lim_{t \to x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x$ for $\forall x \in (0,\pi)$. If $f\left(\frac{\pi}{6}\right) = \frac{-\pi}{4\sqrt{2}}$, then which of the following statement(s) is/are TRUE?

A) $f\left(\frac{\pi}{4}\right) = \frac{-\pi}{4\sqrt{2}}$

B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0,\pi)$

C) There exist $\alpha \in (0,\pi)$ such that $f'(\alpha) = 0$

D) $f''\left(\frac{\pi}{2}\right) - f\left(\frac{\pi}{2}\right) = \pi$



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## SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

**Full Marks :** +4 If only (all) the correct option(s) is (are) chosen.
**Partial Marks:** +3 If all the four options are correct but ONLY three options are chosen.
**Partial Marks:** +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
**Partial Marks :** +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks:** -2 In all other cases.

5. Consider the parabola whose equation is $y = x^2 - 4x$ and the line $y = 2x - b$. Then which of the following is/are CORRECT?

A) For $b = 9$ the line is a tangent to the parabola.

B) For $b = 7$ the line cuts the parabola in A and B such that the $\angle AOB$ is a right angle where O is the origin.

C) For some $b \in R$ the line cuts the parabola in C and D such that x -axis bisects the $\angle COD$ where O is the origin.

D) For $b > 9$ the line passes outside the parabola.

6. If $S = \frac{1}{1+n^2} + \frac{2}{2^2+n^2} + \frac{3}{3^2+n^2} + \ldots\ldots$ Further $S_1$ is the sum of first $(n-1)$ terms of the given series and $S_2$ is the sum of $(n-1)$ terms but starts from the $2^{nd}$ term (i.e. $\frac{2}{2^2+n^2}$) of the given series; $n \in N$ (set of natural numbers). Then which of the following is/are CORRECT?

A) $S_1 > \frac{1}{2}\ln 2$ B) $S_2 < \frac{1}{2}\ln 2$ C) $S_2 > \frac{1}{2}\ln 2$ D) $S_1 < \frac{1}{2}\ln 2$

7. Let $f : R \to R$ be a continuous function satisfying $x^2 - x(f(x)+5) + 6 + 2f(x) = 0; \forall x \in R$. Then which of the following holds good?

A) f is many-one function

B) $f(8) = 5$

C) Area bounded by $y = f(x)$ and co-ordinates axes is $\frac{9}{2}$

D) $f(x) = f^{-1}(x)$ is satisfied for exactly one value of $x$.

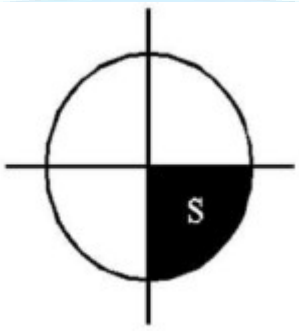


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## SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

8. Let $z_n$ denotes the $n^{th}$ term of a sequence of complex numbers defined by $z_n = (5+12i)\left(\frac{i}{2}\right)^n$. Let S be the intersection of the closed unit disk centered at origin and the fourth quadrant in the complex plane, then the smallest value of 'n' $\in N$ such that $z_n$ lies in S.




9. There is a plane $\alpha x + \beta y + \gamma z + 1 = 0$ which is perpendicular to the plane $2x - y + 2z + 5 = 0$ and containing the line $\vec{r} = (3\lambda - 1)\hat{i} + 6\lambda\hat{j} - 2\lambda\hat{k}$. Then the absolute value of $3\alpha + 2\beta + 4\gamma$ is equal to_____

10. Suppose there exists a point $(x_0, y_0)$ on the circle $(x - r - 1)^2 + y^2 = r^2$ satisfying $4x_0 - y_0^2 \le 0$, then the minimum value of $|r|$ is ____

11. Let N be the number of arrangements of the letters of the word "LOVEISLIFE" in which no two alike letters are together and containing the word LOVE. Then N is_____

12. If the sum of all the solutions of the equation $\left(2\sec^2 x - \cos\frac{16}{9}x\right)\sec x - \left(\cos\frac{x}{3} + 2\tan x \cdot \sec x\right)\tan x = \sin\frac{x}{3}; x \in [-9\pi, 9\pi]$ is $k\pi$, then $2k$ is equal to :



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13. Let N be the number of triples $(A,B,C)$ of sets with union $A\cup B\cup C=\{1,2,3,4,5,6,7,8,9,10\}$ and $A\cap B\cap C=\phi$, then number of even positive integral divisors of N is_____

## SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
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### PARAGRAPH "I"

Let A be matrix of order $3\times3$ with real entries and satisfying the condition

$A^{2019}+A=\begin{pmatrix}2&2&0\\0&2&2\\0&0&2\end{pmatrix}$, then

14. $5tr(A)$ is_____ ($\mathrm{tr(A)}$ is sum the elements in the principal diagonal)

15. $7det(A)$ is_____

### PARAGRAPH "II"

Let $L_1$ and $L_2$ be the following straight lines.

$L_1:\frac{x-1}{1}=\frac{y}{-1}=\frac{z-1}{3}$ and $L_2:\frac{x-1}{-3}=\frac{y}{-1}=\frac{z-1}{1}$. Suppose the straight line

$L:\frac{x-\alpha}{\ell}=\frac{y-1}{m}=\frac{z-\gamma}{-2}$ lies in the plane containing $L_1$ and $L_2$ and passes through the point of intersection of $L_1$ and $L_2$. If the line L bisects the acute angle between the lines $L_1$ and $L_2$, then

16. $\alpha+\gamma=$

17. $\ell\alpha+m\gamma+\ell^2+m^2=$



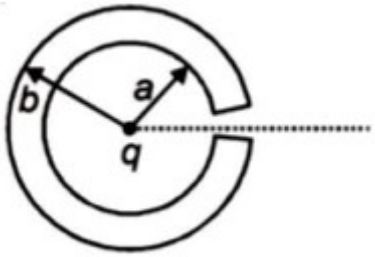
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**PHYSICS** **MAX.MARKS: 60**

## SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
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**Full Marks :** +3 If only the correct option is chosen.
**Zero Marks:** 0 If none of the option is chosen.(i.e the question is un answered)
**Negative Marks:** -1 In all other cases.

18. A point charge $q$ is located at the centre $O$ of a spherical uncharged conducting layer provided with a small orifice. The inside and outside radii of the layer are equal to a and b respectively. What amount of work has to be performed to slowly transfer the charge q from the O through the orifice to infinity?

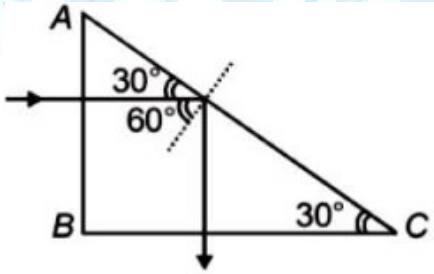



A) $\frac{q^2}{8\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$ B) $\frac{q^2}{4\pi\varepsilon_0}\left(\frac{1}{b}-\frac{1}{a}\right)$ C) $\frac{q^2}{8\pi\varepsilon_0}\left(\frac{1}{b}-\frac{1}{a}\right)$ D) $\frac{q^2}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$

19. A conducting metal circular wire loop of radius r is placed perpendicular to a magnetic field which varies with time as $B = B_0 e^{-\frac{t}{\tau}}$, where $B^0$ and $\tau$ are constant, at time $t = 0$. If the resistance of the loop is $R$, then the heat generated in the loop in a long time $(t \to \infty)$ is

A) $\frac{\pi^2 r^4 B_0^4}{2\tau R}$ B) $\frac{\pi^2 r^4\ B_0^2}{2\tau R}$ C) $\frac{\pi^2 r^4\ B_0^2 R}{\tau}$ D) $\frac{\pi^2 r^4\ B_0^2}{\tau R}$

20. A ray of light is incident normally on a prism of refractive index 1.5, as shown in figure. The prism is immersed in a liquid of refractive index $\mu$. The largest value of the angle ACB, so that the ray is totally reflected at the face AC, is $30^\circ$. The value of $\mu$ must be

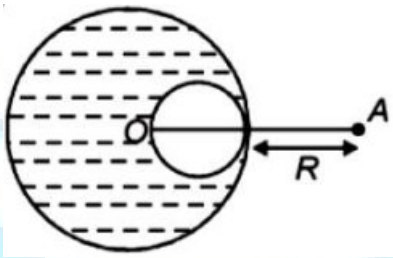



A) $\frac{\sqrt{3}}{2}$ B) $\frac{5}{3}$ C) $\frac{4}{3}$ D) $\frac{3\sqrt{3}}{4}$



For More Join: @IITJEE_Advanced

21. A solid sphere of uniform density and radius R applies a gravitational force of attraction equal to $F_1$ on a particle placed at a distance $2R$ from the centre of the sphere. A spherical cavity at $\frac{R}{2}$ radius is now made in the sphere as shown in figure. The sphere with cavity now applies a gravitational force $F_2$ on the same particle. The ratio $\frac{F_2}{F_1}$ is




A) $\frac{1}{2}$ B) $\frac{3}{4}$ C) $\frac{7}{8}$ D) $\frac{7}{9}$

## SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
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**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks:** -2 In all other cases.

22. A convex lens of focal length 20 cm is placed in front of a convex mirror of focal length 15 cm at a distance of 5 cm. An object of length 2 cm is placed perpendicular to the common axis at a distance of 10 cm from the lens. Then

A) Net magnification after one refraction and one reflection is $\frac{3}{8}$

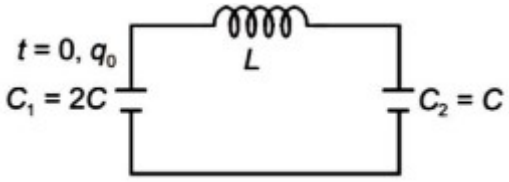
B) Net magnification after one refraction and one reflection is $\frac{3}{4}$

C) Final image (after one refraction and one reflection) is erect (w.r.t object)

D) Final image (after one refraction and one reflection) is inverted (w.r.t object)


For More Join: @IITJEE_Advanced

23. A charged capacitor $C_1$ is connected in series with another uncharged capacitor and an ideal inductor L shown in figure. Initial current in circuit is zero. Choose CORRECT statements




A) The maximum charge on $C_2$ is $\frac{q_0}{3}$

B) The maximum current through inductor is $\frac{q_0}{\sqrt{6LC}}$

C) The maximum current through inductor is $\frac{q_0}{\sqrt{2LC}}$

D) Angular frequency of oscillation is $\sqrt{\frac{3}{2CL}}$

24. A wave disturbance in a medium is described by

$y(x,t)=0.02cos\left(50\pi t+\frac{\pi}{2}\right)cos(10\pi x)$, where $x$ and $y$ are in meters and $t$ in second.

A) A node occurs at $x=0.15\ m$ B) An antinode occurs at $x=0.3\ m$

C) The wave speed is $5\ m/s$ D) The wavelength is 0.2 m

## SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
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Answer to each question will be evaluated according to the following marking scheme:
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25. Consider nuclear reaction: $Lu_{71}^{176}=Hf_{72}^{176}+e^{-}+\bar{v}$.

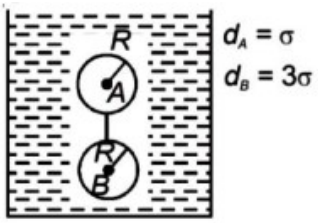
Assume Hf to be at rest.
$Lu_{71}^{176}=175.94269u$ and $Hf_{72}^{176}=175.94142u$ are atomic masses. Maximum KE of $\beta$-particle would be $n\times1.893\times10^{-13}\ J$. Find $n$ $(1u=931.5MeV)$

26. A transverse displacement of a string fixed at both ends is given by

$y(x,t)=0.06sin\frac{2\pi x}{3}cos120\pi t$, where x, y are in meters and t in second. The length of string is 1.5 m and mass $3.0\times10^{-2}$ kg. The tension in string is $n(324)N$. Find $n$

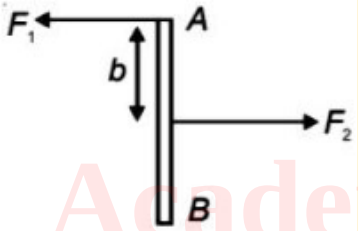

For More Join: @IITJEE_Advanced

27. Two solid spheres are floating in liquid as shown in figure. There is tension in thread connecting them. Find the value of $\frac{\rho}{\sigma}$ (where $\rho$ is density of liquid)




28. A furnace is at a temperature of 2000 K . It emits most intensively at wavelength $x$Å . Find $\frac{x}{2900}$ (Given the Wien's constant $= 2.9 \times 10^{-3} mK$ )

29. A uniform rod AB of mass $m = 1.0\ kg$ moves translationally with acceleration $a = 2.0\ m/s^2$ due to antiparallel forces $F_1$ and $F_2$. The distance between the points at which these forces act is equal to $b = 20\ cm$. Besides it is known that $F_2 = 5.0\ N$. Length of the rod is ____ m.




30. If frequency of $k_\alpha X$ -ray emitted from element with $Z = 31$ is f , then frequency of $k_\alpha - X$ -ray emitted from element with $Z = 51$ would be $\frac{x}{y} f$ . Find $x + y$? $x \& y$ are coprime.

## SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
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For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
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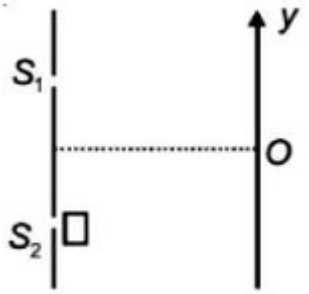
## PARAGRAPH "I"

A Young's double slits experiment is performed in a medium of refractive index $\frac{4}{3}$.

A light of 600 nm wavelength in given medium is falling on the slits having 0.45 mm separation. The lower slit $S_2$ is covered by a thin glass plate of thickness $10.4\mu\ m$ and refractive index 1.5. The interference pattern is observed on a screen placed 1.5 m from slits as shown in figure.



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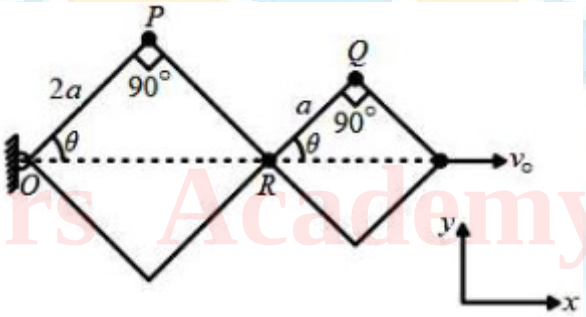



31. The location of the central maxima on the $y$-axis $-\frac{x}{100}\ mm$. Then $x =$ ____

32. The approximate light intensity at point O relative to intensity at maxima is $\frac{y}{4}I_{max}$, then $y =$ ____

## PARAGRAPH "II"

In the given figure, a frame is the combination of two square frames fixed at point ' O ' and rest of all other joints are free to move. For the given instant, ($\theta = 45^\circ$)




33. The magnitude of velocity of point $P$ is $\frac{\sqrt{n}\ V_0}{3}$. Then $n =$ ____

34. y -component of velocity (magnitude) of point Q is $\frac{V_0}{\sqrt{n}}$. Then $n =$ ____

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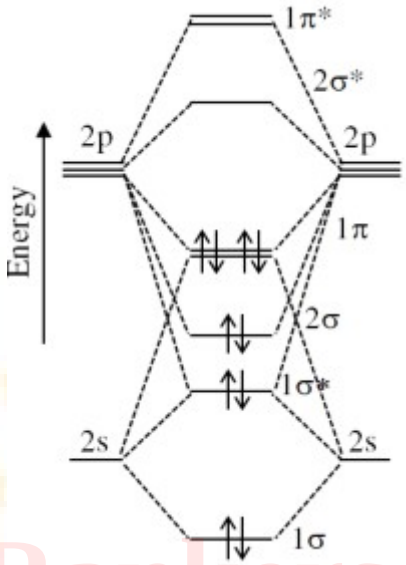
**CHEMISTRY** **MAX.MARKS: 60**

## SECTION–I (Maximum Marks: 12)

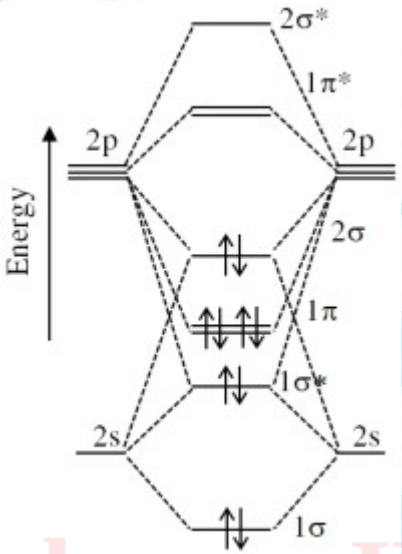
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**Zero Marks:** 0 If none of the option is chosen.(i.e the question is un answered)
**Negative Marks:** -1 In all other cases.

35. The correct molecular orbital diagram of $N_2$ in ground state is :

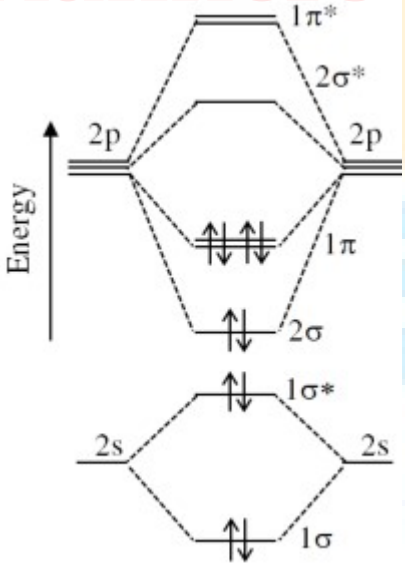
A)



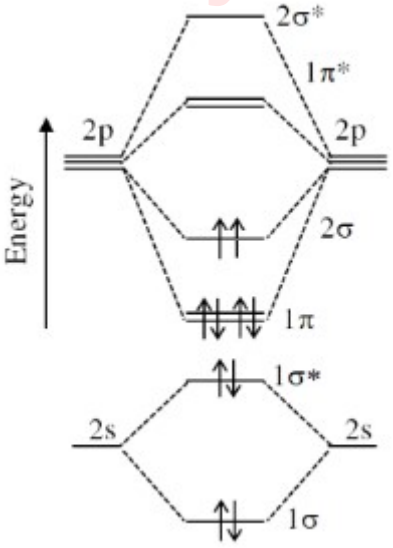

B)




C)



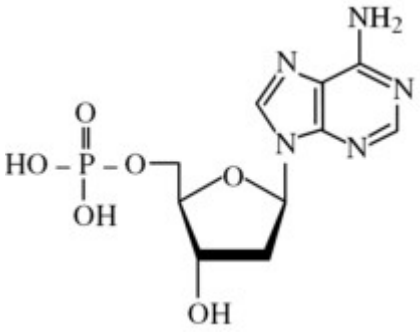

D)

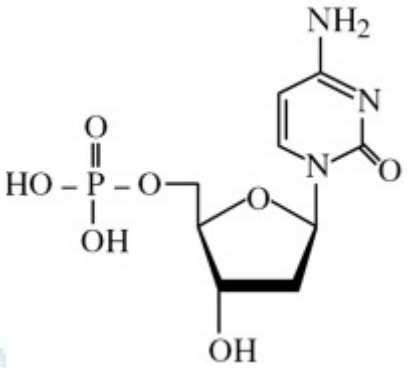


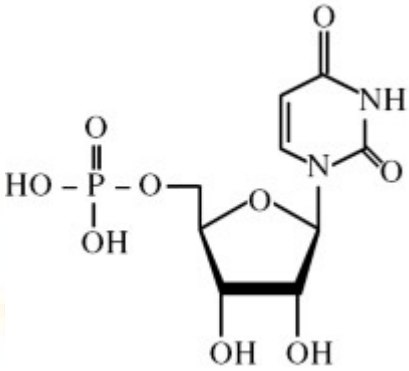


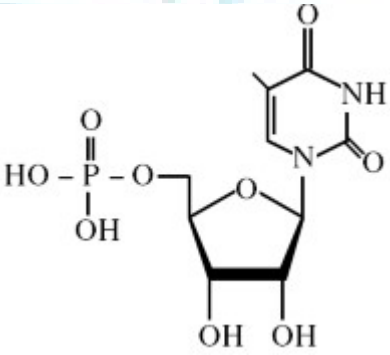

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36. Which of the following is a nucleotide in *m*-RNA?

A) 

B) 

C) 

D) 

37. Consider the following statements related to colloids :

(I) Lyophilic colloids are not formed by simple mixing of dispersed phase and dispersion medium.

(II) Emulsifiers are necessary for the formation of both o/w & w/o type emulsions.

(III) Micelles are produced by dissolving a surfactant in water at any temperature.

(IV) Tyndall effect can be observed from a colloidal solution with dispersed phase having the same refractive index as that of the dispersion medium.

The option with the INCORRECT set of statements is :

A) (I) and (II) B) (II) and (III) C) (I), (III) and (IV) D) (II) and (IV)

38. Which of the following gives a chiral product on reductive ozonolysis?

A) cis-1, 4-polyisoprene B) trans-1, 4-polyisoprene

C) Neoprene D) Butadiene-styrene copolymer



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## SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

**Full Marks :** +4 If only (all) the correct option(s) is (are) chosen.
**Partial Marks:** +3 If all the four options are correct but ONLY three options are chosen.
**Partial Marks:** +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
**Partial Marks :** +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
**Zero Marks :** 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks:** -2 In all other cases.

39. In this question, each option has two statements, **S-I** and **S-II**. **S-II** is a statement related to **S-I**. Choose the option(s) in which S-II is CORRECT explanation of **S-I**.

A) **S-I:** Ethyl amine is soluble in water whereas aniline is not.

**S-II:** The hydrophobic part of aniline is larger than that of ethyl amine.

B) **S-I:** Methyl amine in water reacts with ferric chloride to precipitate hydrated ferric oxide.

**S-II:** Methylamine acts as a Bronsted base in water.

C) **S-I:** Although amino group is o- and p- directing in aromatic electrophilic substitution reactions aniline on nitration gives a substantial amount of *m*-nitroaniline.

**S-II:** Aniline undergoes protonation in acidic medium to become anilinium ion.

D) **S-I:** $Me_3N$ is more basic than $Me_2NH$ in vapour phase while reverse is true in aqueous solution.

**S-II:** Inductive effect dominates over hydrogen bonding in aqueous solution.

40. The first ($\Delta_r H_1$) and the second ($\Delta_r H_2$) ionization enthalpies (in $kJmol^{-1}$) and the ($\Delta_{eg} H$) electron gain enthalpy (in $kJmol^{-1}$) of a few elements are given below:

| Elements | $\Delta_r H_1$ | $\Delta_r H_2$ | $\Delta_{eg} H$ |
|---|---|---|---|
| I | 520 | 7300 | -60 |
| II | 419 | 3051 | -48 |
| III | 1681 | 3374 | -328 |
| IV | 1008 | 1846 | -295 |
| V | 2372 | 5251 | +48 |
| VI | 738 | 1451 | -40 |

Choose the CORRECT options(s):


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A) The most reactive metal is II.

B) III and IV are both non-metals, III being the most reactive non-metal.

C) VI is a metal, and it can form a stable binary halide of the formula MX (X = halogen).

D) **V** is the most unreactive element.

41. A hydrocarbon **X** on oxidative ozonolysis gives **Y** as the only product. 0.1 mol of **Y** when dissolved in 1000 g of benzene lowers the freezing point of benzene by $0.25^{\circ}C$. The solution gives 12.2 g solid residue on complete evaporation of benzene. Cryoscopic constant of benzene is $5\ K\ kg\ mol^{-1}$. Choose the correct statement(s) from the following:

A) **Y** is 100% dimerized in benzene

B) The molecular formula of **Y** may be $C_7H_6O_2$

C) **X** exhibits optical isomerism.

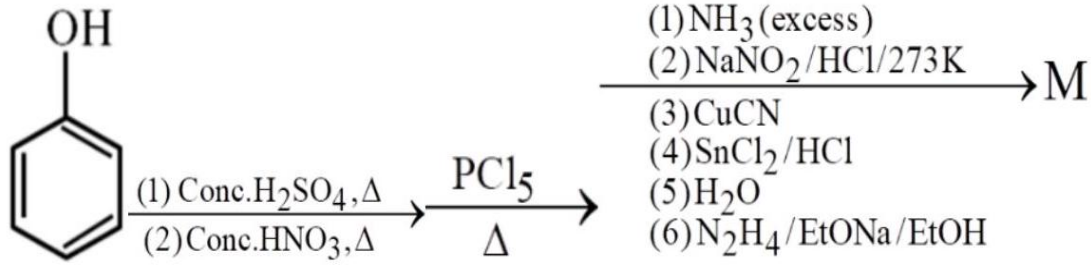
D) Degree of unsaturation of **X** may be 9

## SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +4 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

42. The sum of the number of amines of the formula $C_5H_{13}N$ excluding stereoisomers is:

43. Consider the following reaction sequence:




The sum of number heteroatoms &$sp^2$ carbon atoms in a molecule of **M** is ____ .

44. Among $[I_3]^-, XeF_2, ClF_3, XeOF_2, PF_5, Fe(CO)_5, [PCl_4]^+, IF_5$ & $[XeF_3]^+$, the number of species having $sp^3d$ or $dsp^3$ hybridized central atom is ____

45. $H_2S(g)$ is bubbled through a solution having 1 mole of $KMnO_4$ in an acidic solution till it gets completely decolorized giving X g of a precipitate. Similarly, 1 mole of $K_2Cr_2O_7$ gives Yg of the same precipitate. The sum of $(X+Y)$ is ____

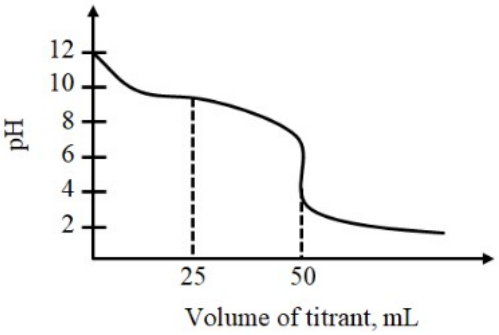
[Atomic masses: $H=1, O=16, K=39, Mn=55, Cr=52, S=32$]



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46. 1 mol of $SO_2Cl_2(g)$ is heated to $227^\circ C$ so that the following reaction occurs: $SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g); \Delta H = +20 kcal.mol^{-1}$ If the specific rate of the reaction is $0.2303\ min^{-1}$, find the loss of entropy of the surrounding in cal. $K^{-1}$ by the end of 10 minutes.

47. The following titration curve was obtained as part of a general chemistry laboratory experiment for an unknown substances, that weighed 1.50 g . The titrant was either a $0.3M$ strong base or $0.3M$ strong acid. Estimate the molar mass of the unknown.




## SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +3 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

### "PARAGRAPH- I"

In a hydrogen like species, the electron is in the $2^{nd}$ excited state and has an energy of -13.6 eV. The energy required to excite the electron to $4^{th}$ energy level is $\mathbf{x}eV$ and the ionization energy of the species is $\mathbf{y}eV$ .

48. The value of **x** is ____ .

49. The value of $y$ is ____ .

### "PARAGRAPH- II"

59.5 g a metallic tin reacts with excess conc. HCl to give '**x**g' of a salt **P**. Just sufficient amount of 2,4,6-trichlorobenzonitrile is added to this solution so that the salt reacts with it completely. After the reaction is over, excess water is added to the mixture and distilled to get '**y**g' of an organic compound.

$[H = 1, C = 12, N = 14, O = 16, Cl = 35.5, Sn = 119]$

50. The value of **x** is ____ .

51. The value of $y$ is ____ .



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